MING-CHIN LIN, MD, PhD

Attending Neurosurgeon | Disorders of Consciousness Research | AI-Based Neurological Prognosis Prediction

Current Positions

• Attending Neurosurgeon, Department of Neurosurgery, Wan Fang Hospital, Taipei Medical University
• Associate Professor, Graduate Institute of Biomedical Informatics, Taipei Medical University
• Vice President, International Medical Informatics Association (IMIA), 2025–2027
• President, Taiwan Association for Medical Informatics

Education

• Ph.D. in Biomedical Informatics, University of Utah, USA
• M.D., School of Medicine, Taipei Medical University
• Board-Certified Neurosurgeon

Research Experience Relevant to This Lecture

[Disorders of Consciousness and Neurological Prognosis Prediction]

• Principal Investigator, multicenter project on multimodal machine-learning prediction of brain-injury prognosis
  - Collaborating hospitals: Shuang Ho Hospital, National Taiwan University Hospital, Wan Fang Hospital, Hualien Tzu Chi Hospital, Hsiao Chung-Cheng Healthcare System
  - Over 800 brain-injury patients enrolled
  - Patient types: traumatic intracerebral hemorrhage, spontaneous intracerebral hemorrhage, cerebral infarction, hypoxic brain injury, etc.

• Developed the digital biomarker “Brain Connectivity Entropy”
  - Quantifies the complexity and integration of functional brain networks
  - Overcomes the reliance of conventional behavioral assessment on motor function
  - Objectively evaluates recovery potential in patients with disorders of consciousness

• Established Taiwan’s largest task-based EEG database
  - Integrates multimodal data including EEG signals, physiological monitoring, electronic health records, and neuroimaging

• Developed the AI Brain Response Clinical Decision Support System (BRes-CDSS)
  - AUC of 0.872 for neurological prognosis prediction

• Developed an AIoT wearable wireless EEG monitoring system
  - Second-generation Bluetooth EEG device (medical device certification submitted)

Neurosurgical Clinical Expertise

• ROSA Robotic-Arm-Assisted Surgery
  - 350+ ROSA robotic-arm procedures performed
  - Instructor, Asia-Pacific ROSA Training Center
  - Robotic-arm-assisted intracerebral hemorrhage aspiration

• Stereotactic and image-guided surgery
• Gamma Knife radiosurgery

Selected Publications

1. Luh HT, et al., Lin MC, Huang APH. Application of Robotic Stereotactic Assistance (ROSA) for spontaneous intracerebral hematoma aspiration and thrombolytic catheter placement. Journal of the Formosan Medical Association. 2025;124(5):452-461.

2. Ng BMK, Liu HW, Su YK, Lo WL, Lin CM. Automatic Trajectory Planning for Robotic Arm in Putaminal Hemorrhage Aspiration. Studies in Health Technology and Informatics (MEDINFO 2025). 2025;329:1862-1863.

3. Dewi SRK, Li YCJ, Lin MC. Brain Response and 6-Month Responsiveness Exploration in Brain Infarction Case: Preliminary Study. Studies in Health Technology and Informatics. 2025;329:1870-1871.

4. Heath DC, et al., Lin MC, et al. Comparison between robot-assisted and navigation-guided minimally invasive transforaminal lumbar interbody fusion: a multicenter study. Neurosurgical Focus. 2024;57(6):E12.

Research Infrastructure

Smart Medical Information Mart for Neuro Intensive Care (MiNiC)
An intelligent informatics platform for neurocritical care: a multimodal neurocritical-care database integrating task-based EEG, physiological signals, and clinical records to support AI-based neurological prognosis prediction research.